

$$\begin{aligned}
&= P_k^+ H_k^T R_k^{-1} \\
\hat{x}_k^- &= F_{k-1} \hat{x}_{k-1}^+ + G_{k-1} u_{k-1} = \text{a priori state estimate} \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-) = \text{a posteriori state estimate} \\
P_k^+ &= (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k R_k K_k^T \\
&= [(P_k^-)^{-1} + H_k^T R_k^{-1} H_k]^{-1} \\
&= (I - K_k H_k) P_k^- \tag{5.19}
\end{aligned}$$

The first expression for P_k^+ above is called the Joseph stabilized version of the covariance measurement update equation. It was formulated by Peter Joseph in the 1960s and can be shown to be more stable and robust than the third expression for P_k^+ [Buc68, Cra04] (see Problem 5.2). The first expression for P_k^+ guarantees that P_k^+ will always be symmetric positive definite, as long as P_k^- is symmetric positive definite. The third expression for P_k^+ is computationally simpler than the first expression, but its form does not guarantee symmetry or positive definiteness for P_k^+ . The second form for P_k^+ is rarely implemented as written above but will be useful in our derivation of the information filter in Section 6.2.

If the second expression for K_k is used, then the second expression for P_k^+ must be used. This is because the second expression for K_k depends on P_k^+ , so we need to use an expression for P_k^+ that does not depend on K_k .

Note that if x_k is a constant, then $F_k = I$, $Q_k = 0$, and $u_k = 0$. In this case, the Kalman filter of Equation (5.19) reduces to the recursive least squares algorithm for the estimation of a constant vector as given in Equation (3.47).

Finally we mention one more important practical aspect of the Kalman filter. We see from Equation (5.19) that the calculation of P_k^- , K_k , and P_k^+ does not depend on the measurements y_k , but depends only on the system parameters F_k , H_k , Q_k , and R_k . That means that the Kalman gain K_k can be calculated offline before the system operates and saved in memory. Then when it comes time to operate the system in real time, only the $\hat{x}_k$ equations need to be implemented in real time. The computational effort of calculating K_k can be saved during real-time operation by precomputing it. If the Kalman filter is implemented in an embedded system with strict computational requirements, this can make the difference between whether or not the system can be implemented in real time. Furthermore, the performance of the filter can be investigated and evaluated before the filter is actually run. This is because P_k indicates the estimation accuracy of the filter, and it can be computed offline since it does not depend on the measurements. In contrast, as we will see in Chapter 13, the filter gain and covariance for nonlinear systems cannot (in general) be computed offline because they depend on the measurements.

5.2 KALMAN FILTER PROPERTIES

In this section, we summarize some of the interesting and important properties of the Kalman filter. Suppose we are given the linear system of Equation (5.17) and we want to find a causal filter that results in a state estimate $\hat{x}_k$. The error between the true state and the estimated state is denoted as $\tilde{x}_k$:

$$\tilde{x}_k = x_k - \hat{x}_k \tag{5.20}$$

Since the state is partly determined by the stochastic process $\{w_k\}$, x_k is a random variable. Since the state estimate is determined by the measurement sequence $\{y_k\}$, which in turn is partly determined by the stochastic process $\{v_k\}$, $\hat{x}_k$ is a random variable. Therefore, $\tilde{x}_k$ is also a random variable.

Suppose we want to find the estimator that minimizes (at each time step) a weighted two-norm of the expected value of the estimation error $\tilde{x}_k$:

$$\min E [\tilde{x}_k^T S_k \tilde{x}_k] \quad (5.21)$$

where S_k is a positive definite user-defined weighting matrix. If S_k is diagonal with elements $S_k(1), \dots, S_k(n)$, then the weighted sum is equal to $S_k(1)E[\tilde{x}_k^2(1)] + \dots + S_k(n)E[\tilde{x}_k^2(n)]$.

- If $\{w_k\}$ and $\{v_k\}$ are Gaussian, zero-mean, uncorrelated, and white, then the Kalman filter is the solution to the above problem.
- If $\{w_k\}$ and $\{v_k\}$ are zero-mean, uncorrelated, and white, then the Kalman filter is the best linear solution to the above problem. That is, the Kalman filter is the best filter that is a linear combination of the measurements. There may be a nonlinear filter that gives a better solution, but the Kalman filter is the best linear filter. It is often asserted in books and papers that the Kalman filter is not optimal unless the noise is Gaussian. However, as our derivation in this chapter has shown, that is simply untrue. Such statements arise from erroneous interpretations of Kalman filter derivations. Even if the noise is not Gaussian, the Kalman filter is still the optimal *linear* filter.
- If $\{w_k\}$ and $\{v_k\}$ are correlated or colored, then the Kalman filter can be modified to solve the above problem. This will be shown in Chapter 7.
- For nonlinear systems, various formulations of nonlinear Kalman filters approximate the solution to the above problem. This will be discussed further in Chapters 13–15.

Recall the measurement update equation from Equation (5.19):

$$\hat{x}_k^+ = \hat{x}_k^- + K_k(y_k - H_k \hat{x}_k^-) \quad (5.22)$$

The quantity $(y_k - H_k \hat{x}_k^-)$ is called the innovations. This is the part of the measurement that contains new information about the state. In Section 10.1, we will prove that the innovations is zero-mean and white with covariance $(H_k P_k^- H_k^T + R_k)$. In fact, the Kalman filter can actually be derived as a filter that whitens the measurement and hence extracts the maximum possible amount of information from the measurement. This was first proposed in [Kai68]. When a Kalman filter is used for state estimation, the innovations can be measured and its mean and covariance can be approximated using statistical methods. If the mean and covariance of the innovations are not as expected, that means something is wrong with the filter. Perhaps the assumed system model is incorrect, or the assumed noise statistics are incorrect. This can be used in real time to verify Kalman filter performance and parameters, and even to adjust Kalman filter parameters in order to improve performance. An application of this idea will be explored in Section 10.2.